The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL THREE MODULE, SPRING SEMESTER 2021-2022

Computer Vision

Time allowed: TWO HOURS

This is an open-book shorter-timed examination. Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced

Answer ALL THREE Questions in Section A. Answer THREE out of FOUR Questions in Section B.

ADDITIONAL MATERIAL: Nil.

INSTRUCTION FOR STUDENTS:

- The university's honour code applies to this individual short time assessment. The submitted answers must be your own work where answers are attempted without any help from others. University academic misconduct rules apply here in the case of plagiarism, collusion or false authorship.
- Write the name and student ID at the top of your answer sheets.
- Students are not allowed to ask questions during the short time assessment in accordance with the no shout-out policy.
- You may answer the questions in any order but write the question numbers correctly and clearly.
- At the end of the exam, upload your solutions and name the file as StudentID. Upload your solution file onto Moodle.
- Please ensure that you upload the correct files to the folder labelled as "Final examination submission folder" in Moodle. Request for resubmission will not be accepted after the submission.
- Students should stop writing the answers at the end of exam time. Answer scripts must be submitted to Moodle within thirty minutes after the end of exam. In case of any technical issues, the students need to contact the module convener in this period.

IMPORTANT: Late submission will not be accepted.

SECTION A

Answer ALL THREE Questions in This Section.

Question 1. The following questions are pertaining to Image Features.

[**overall 13 marks**]

a. Determine the Local Binary Pattern (LBP) for the central pixel in the image patch in Figure 1. Explain in general how LBP features are further organised to produce image features that can be used for downstream tasks, e.g., image classification.

90	95	91
85	92	95
90	100	98

Figure 1

[5 Marks]

b. Discuss how the four steps of SIFT, i.e., scale-space extrema detection, key-point localisation, orientation assignment, and generating key-point descriptor, are reflected in the set of SIFT features of a toy car image in Figure 2.

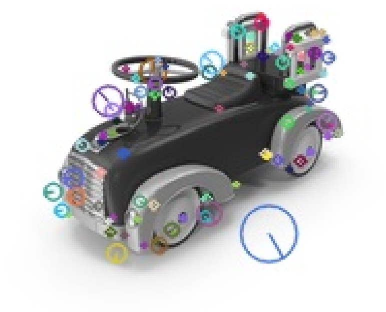


Figure 2

[8 Marks]

Question 2. The following questions are pertaining to Projective Geometry and Camera Geometry.

[**overall 13 marks**]

a. Assume $P = (3, 4, 5)^T$ is a homogeneous representation of a point. Is it a 2-dimensional (2D) point or a 3-dimensional (3D) point? What is its corresponding inhomogeneous representation? Given another point with its homogeneous representation as $Q = (2, 1, 3)^T$, derive the homogeneous representation of the line determined by the two points P and Q.

[6 Marks]

b. A pinhole camera has a focal length of 26mm, 72dpi (i.e., 28.35 pixels per cm) for both horizontal and vertical resolution, and 4032 pixels horizontally and 2268 pixels vertically. Assuming there is no distortion in the projection, calculate the intrinsic parameter matrix. Show your work.

[7 Marks]

Question 3. The following questions are pertaining to Convolutional Neural Networks.

[**overall 14 marks**]

a. A one-channel input feature map of size 3×3 is shown in Figure 3(a). Two filters of size 2×2 are shown in Figure 3(b). Assuming no zero-padding, what is the size of the output feature map based on the 2-dimensional (2D) convolutional operation with stride 1? Calculate the output feature value at the upper-left corner of the output feature maps. Show your work.

90	95	91
85	92	95
90	100	98

(a)

-1	1
2	4

3	-2
-1	-1

(b)

Figure 3

[7 Marks]

b. Given an input RGB image of size 256×256, design a simple Convolutional Neural Network (CNN) that consists of three trainable layers to output a feature map of size 64×64 with 256 channels. You may only choose from operations of 3×3 convolution, 1×1 convolution, and/or 2×2 max pooling. Explain your design.

[7 Marks]

SECTION B

Answer THREE Out of FOUR Questions in This Section.

Question 4: The following questions are pertaining to Image Segmentation.

[**overall 20 marks**]

a. A 3×3 image patch is shown in Figure 4. Given an initial guess of the cluster centers $a = 100, b = 150$, using k-means algorithm determine the label for each image pixel after the cluster centres are updated once. Show your work.

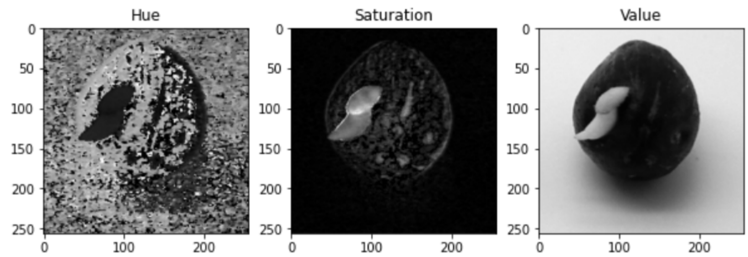
90	95	91
85	120	150
50	200	180

Figure 4

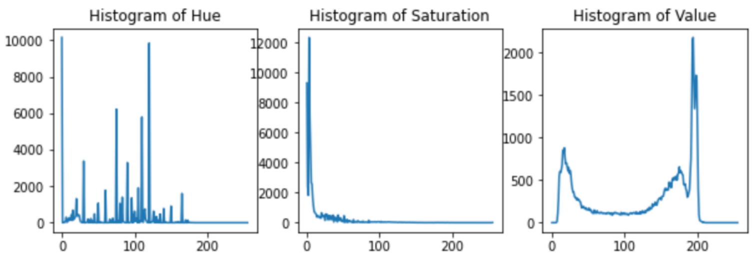
[10 Marks]

b. The anatomy of a germinated oil palm seed is shown in Figure 5. The Hue, Saturation, and Value (HSV) channels of the oil palm seed image are shown in Figure 6a, and the histograms of the three channels are shown in Figure 6b, respectively. Either k-means or a Gaussian Mixture Model (GMM) may be used to segment the image into three regions, i.e., seed body, plumule and radicle, and background, based on the HSV representation of the image. Compare and contrast the advantages and disadvantages of using k-means and a Gaussain Mixture Model (GMM) for the given scenario.

Figure 5



(a) HSV



(b) Histograms of HSV channels

Figure 6

[10 Marks]

Question 5: The following questions are pertaining to Stereo Vision and Multi-View Stereo

[**overall 20 marks**]

a. A pair of rectified stereo images are shown in Figure 7a and 7b respectively. The pixel at the centre of the black circle in the left image corresponds to the pixel at the centre of the black circle in the right image. The pixel coordinates are $(58, 22)$ for the left and $(109, 22)$ for the right, respectively. Similarly, the pixel at the centre of the white circle in the left image corresponds to the pixel at the centre of the white circle in the right image. The pixel coordinates are $(68, 45)$ for the left and $(117, 45)$ for the right, respectively. Calculate the disparity gradient between the two pixels at the centre of the black and white circle. Based on your result, explain whether it violates the disparity gradient limit.

(a) Left image (b) Right image

Figure 7

[10 Marks]

b. Explain the concept of Plane Sweep Stereo and how it is used to reconstruct the depth map of a 3D scene from multiple-view images. Does the Plane Sweep algorithm require known camera parameters? Does it require images to be rectified? Explain.

[10 Marks]

Question 6: The following questions are pertaining to Object Recognition.

[**overall 20 marks**]

a. The Viola-Jones method relies on efficient integral images to generate a large amount of simple image features. Given the 3×3 image patch shown in Figure 8, calculate its integral image, and use this to calculate the simple feature based on subtracting the light-gray region from the dark-gray region. Explain how such pattern can be used to generate multiple simple features in the given image patch.

90	95	91
85	120	100
50	70	60

Figure 8

[10 Marks]

b. Both pictorial structure and the bag-of-feature method can be used to learn a face detector. Briefly explain the concept of a pictorial structure model and bag-of-features method for face detection, respectively. Assume a pictorial model and a bag-of-features model are learned from a set of human face images. Will the pictorial structure model succeed in detecting a human face in the image in Figure 9? Will the bag-of-features model succeed? Explain.

Figure 9

[10 Marks]

Question 7: The following questions are pertaining to Motion and Optical Flow.

[**overall 20 marks**]

a. Figure 10 shows an opaque square object (gray colour) is moving to the top-right. Explain if the motion at three apertures (a), (b), and (c) can be fully recovered. For the ones that cannot be fully recovered, explain which components of the motion could be recovered, if any. Justify your answer.

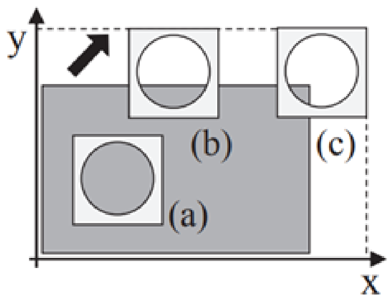


Figure 10

[6 Marks]

b. Two 5×5 image patches at time t and $t+1$ are given in Figure 11(a) and (b), respectively. Calculate the image gradients along x, y, and the time t directions for a 3×3 window centered at the central pixel of the image patch in (a). Assuming the origin of image coordinate system is at the upper-left corner of the image, use the formula $I_x = I(x+1, y) - I(x, y)$ and $I_y = I(x, y+1) - I(x, y)$ to calculate the image gradients along x, y axis, respectively. Based on Lucas-kanade method, explain how these image gradients may be combined to estimate the optical flow vector of the pixel at the centre of the image patch in (a). There is no need to calculate the optical flow vector.

215	214	214	214	214
215	214	214	214	214
215	215	215	215	214
215	215	215	215	213
215	214	214	214	213

(a)

201	201	200	201	212
201	201	196	196	207
201	200	195	195	200
201	198	192	193	193
201	196	196	189	191

(b)

Figure 11

[14 Marks]